

Balaji Ramachandran

Mathematics PhD Student · Carnegie Mellon University

[Academic website](#) [Google Scholar](#) [LinkedIn](#)

RESEARCH INTERESTS

Stochastic analysis and partial differential equations; generative diffusion modelling; high-dimensional and multimodal sampling; reinforcement learning; mathematical finance; stochastic games, control, and optimization.

EDUCATION

Carnegie Mellon University

June 2026 – present

PhD student, Mathematics

Research direction: mathematical finance.

Indian Institute of Science

From 2022

BTech, Mathematics and Computer Science

Bengaluru, India

PUBLICATIONS

A Network Formation Game for Katz Centrality Maximization: A Resource Allocation Perspective

Balaji R, Prashil Wankhede, Pavankumar Tallapragada.

Accepted at the 65th IEEE Conference on Decision and Control (CDC), 2026. Honolulu, Hawaii, December 15–18.

Presenting in person.

arXiv:2604.03056 · doi:10.48550/arXiv.2604.03056

RESEARCH AND PROFESSIONAL EXPERIENCE

HSBC

2026

Quantitative analyst intern

Worked on stress-scenario risk management dashboards and portfolio risk analysis for the FRTB team.

Purdue University

June – August 2025

Graduate research intern

Developed a physics-informed neural network solution for a PDE to estimate diffusion hitting times. Worked on nonparametric impulse control of diffusion processes.